



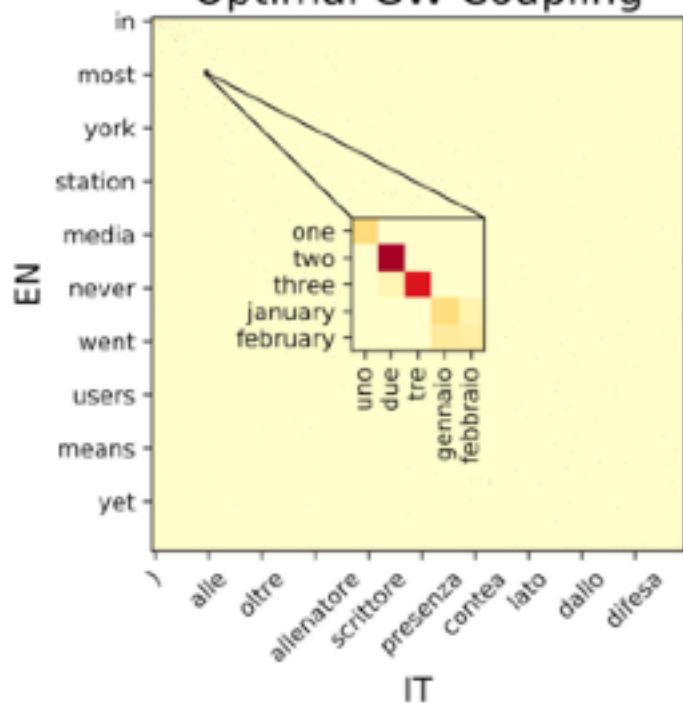


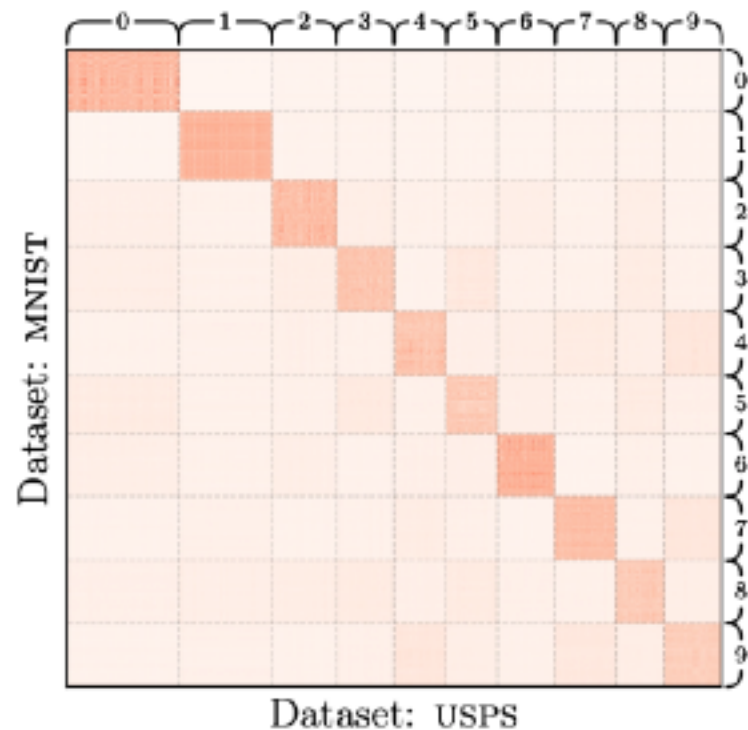
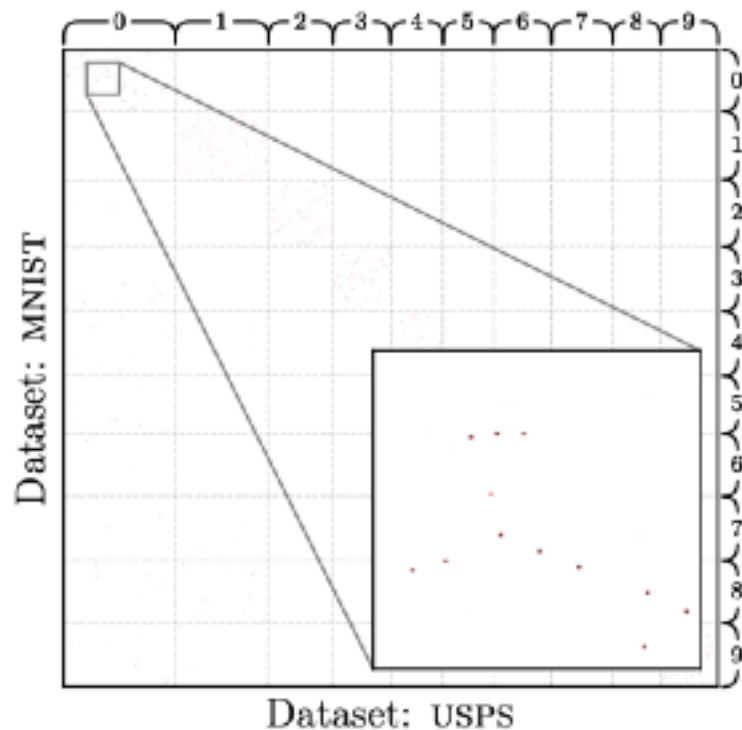
**Coupling tells us “which source points map to which target points**

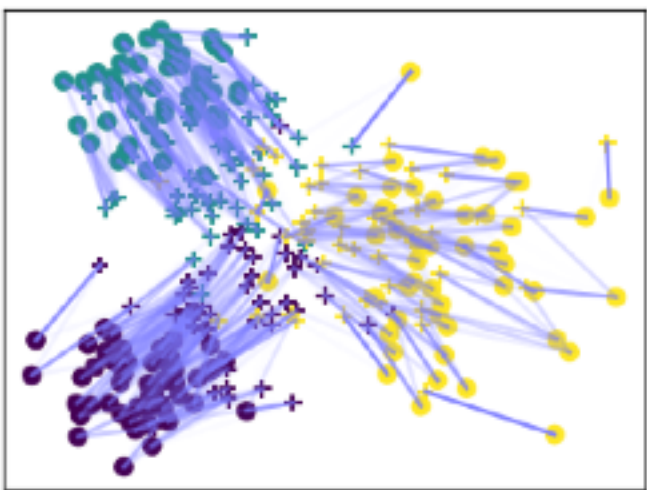
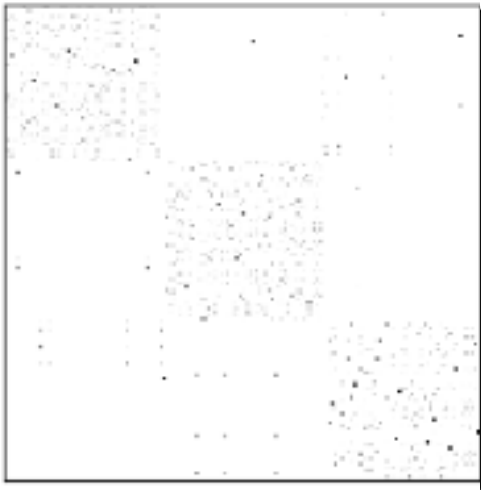
**Couplings capture soft correspondences between vocabularies**

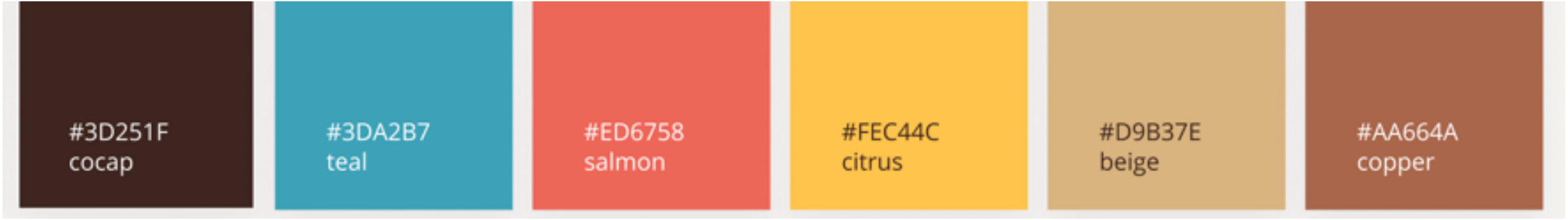
**Useful when multiple plausible correspondences exist**

# Optimal GW Coupling



Optimal Coupling ( $\varepsilon = 0.1$ )Optimal Coupling ( $\varepsilon = 0.0001$ )





#3D251F  
cocap

#3DA2B7  
teal

#ED6758  
salmon

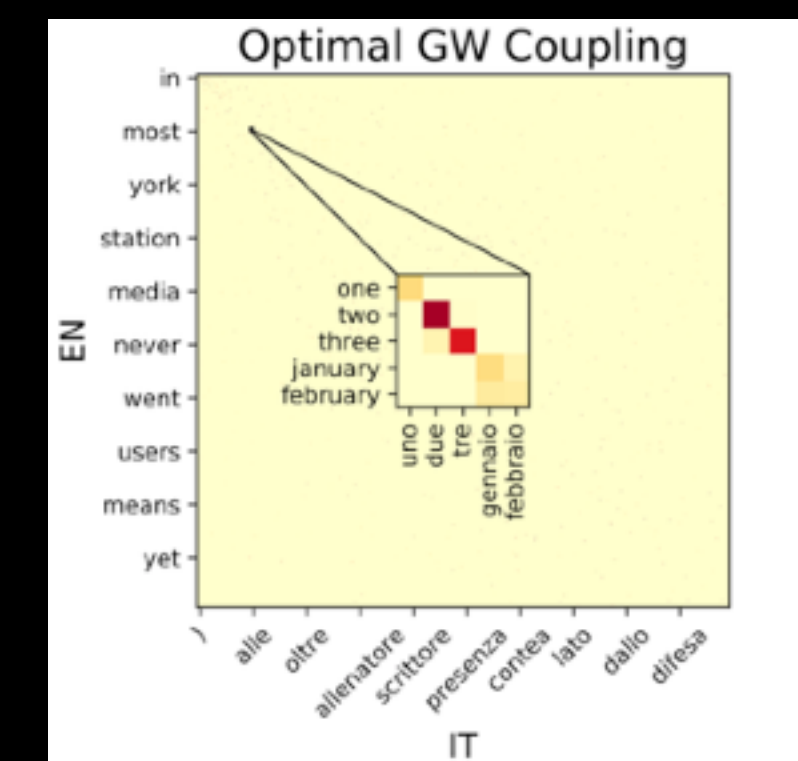
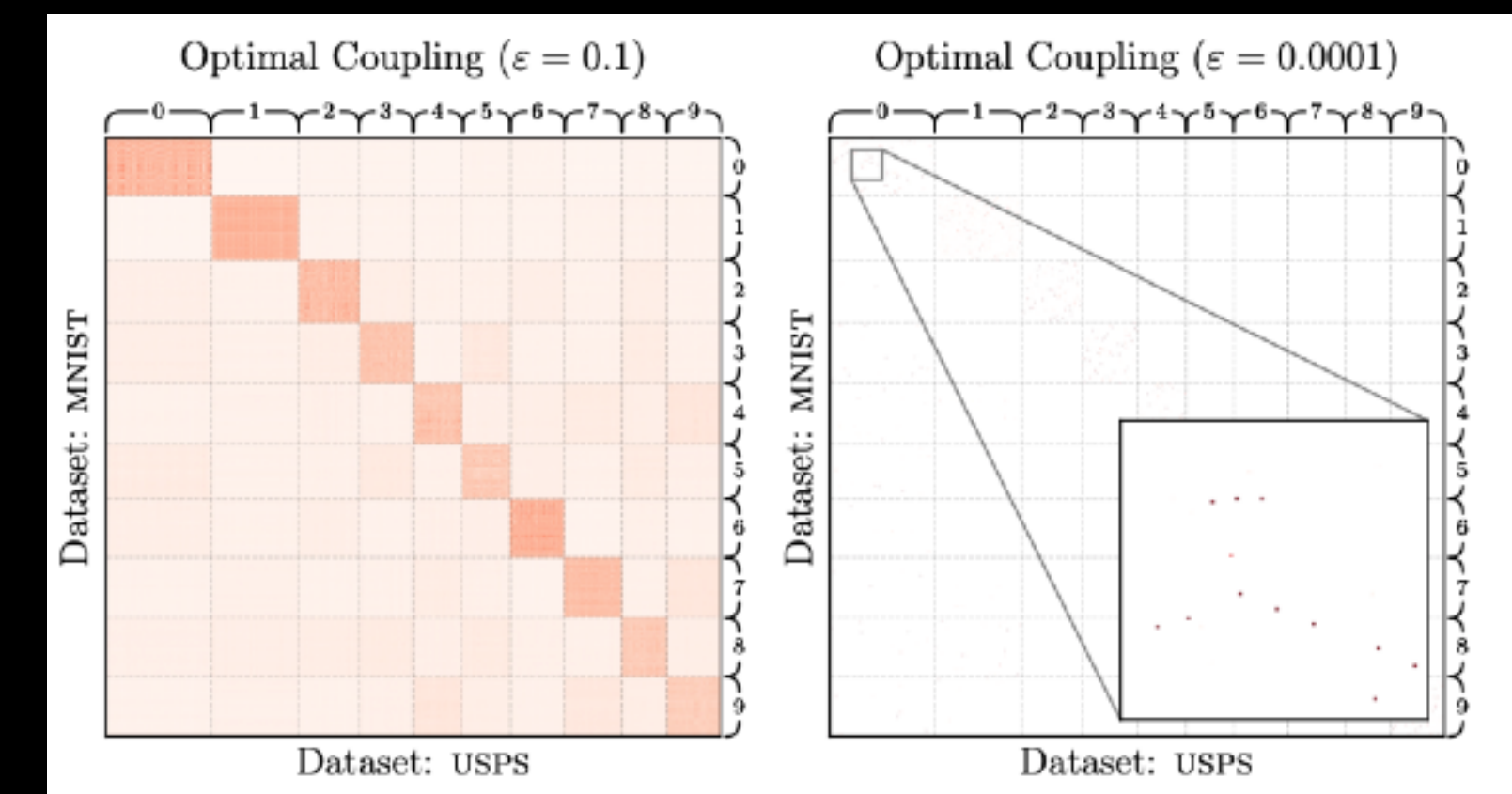
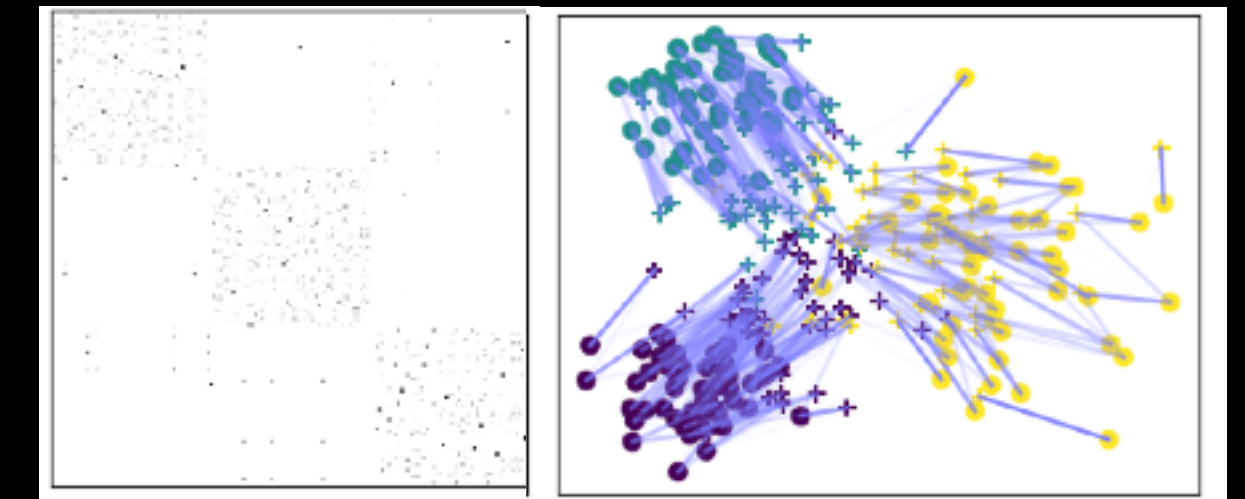
#FEC44C  
citrus

#D9B37E  
beige

#AA664A  
copper

# ML Applications of Kantorovich

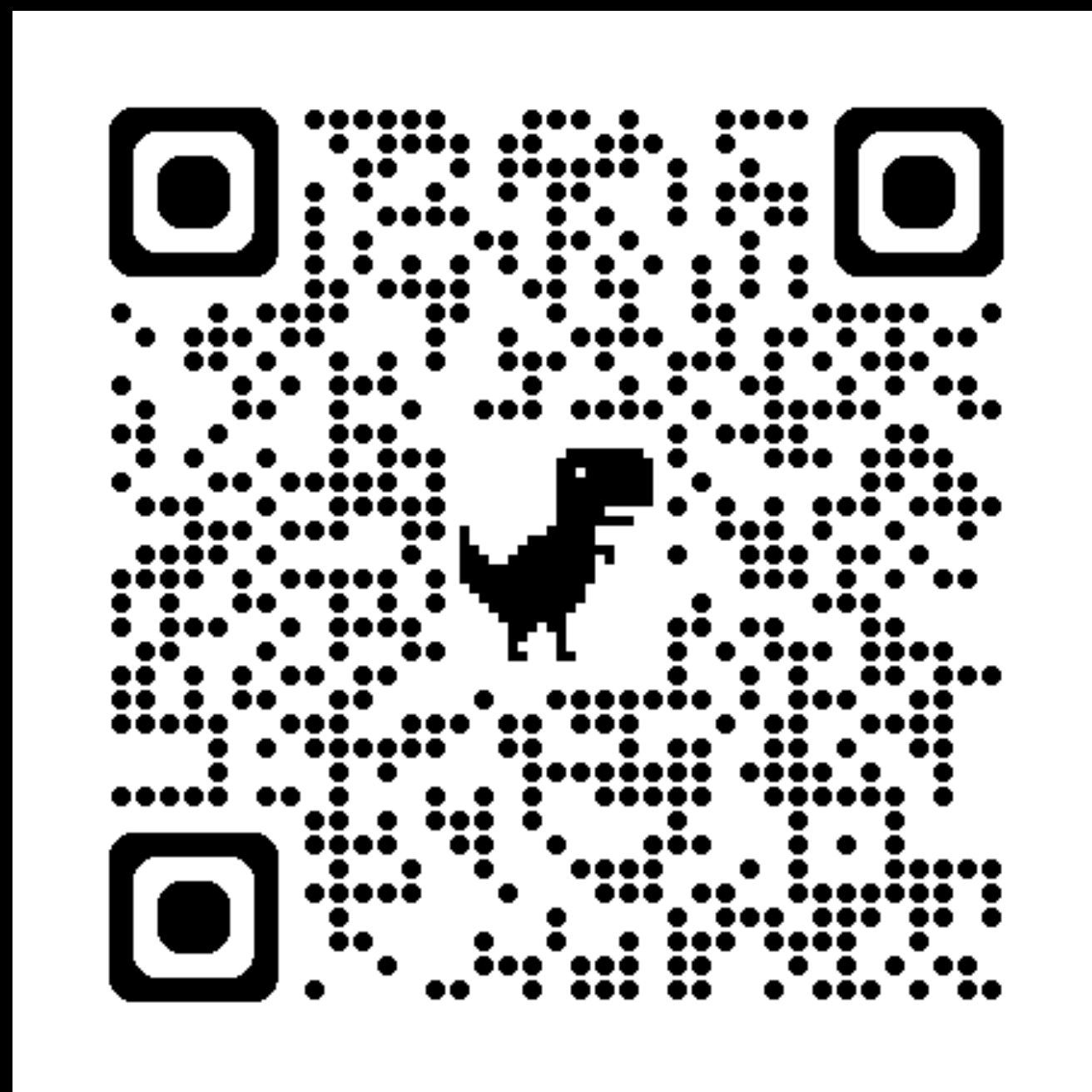
- Domain adaptation: matching datasets
  - Align source and target domains when marginal distributions differ.
  - Example: reweight source training samples to match target distribution.
  - **Coupling tells us “which source points map to which target points”** (probabilistically).
- Embedding alignment
  - Match vector spaces across languages or modalities.
  - Example: bilingual word embeddings (align English & Italian spaces without parallel data).
  - **Couplings capture soft correspondences between vocabularies.**
- Uncertainty-aware mappings via probabilistic transport
  - Unlike Monge maps, couplings allow splitting mass  $\rightarrow$  captures uncertainty.
  - **Useful when multiple plausible correspondences exist** (e.g., many-to-many)
  - Applications: shape matching under noise, multimodal prediction.



# Interactive Demo: Kantorovich

Implement code to:

1. Turn  $\mathbf{a}$ ,  $\mathbf{b}$ ,  $\mathbf{C}$  into LP objective and constraints for an LP solver
2. Recover  $\pi$  from LP solution and compare to Monge solution



Standard form: